8

Determinants

Historical Note

Determinants are older than matrices by a century and a half. Seki Takakazu described them in Japan in 1683, and Leibniz independently wrote one down in a letter to l'Hôpital in 1693, in both cases as the condition for a system of equations to be solvable. Cramer published his rule in 1750, Vandermonde in 1771 was the first to study determinants as objects worth attention in their own right rather than as a device for solving systems, and Cauchy gave the systematic theory in 1812, borrowing for the array a word Gauss had used for the discriminant of a quadratic form. Only in 1850 did Sylvester coin the word *matrix*, and he coined it, as we noted in Chapter 4, for the rectangular womb *out of which determinants could be cut*. The array was invented to hold the determinant, not the other way round. This chapter inverts the history one more time, defining the determinant by the properties we want it to have and then proving that exactly one function has them.

Chapter 7 left a promissory note: it claimed that the determinant is a property of an operator and not merely of an array, and deferred the proof. It also left a gap in the geometry. We have measured length and angle since Chapter 2 but never area or volume, and a linear map plainly does something definite to volume, since it carries the unit cube to a parallelepiped. The determinant is the answer to both questions at once, and the second one is the better way to remember it.

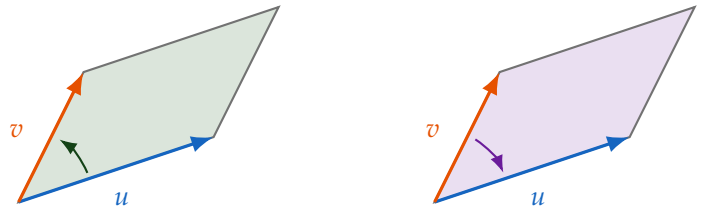
8.1 What we are looking for

Take $n = 2$. The columns $u = (a, c)$ and $v = (b, d)$ span a parallelogram, and elementary geometry gives its area as $|ad - bc|$. Three features of that expression are worth noticing before we generalise, because they are the only features we shall use.

- **It scales.** Doubling u doubles the area, and the expression is linear in a and c with b, d held fixed. The same holds in the other column. It is *not* linear in the pair jointly: doubling both quadruples the area.

- **It vanishes when the columns coincide.** If $u = v$ the parallelogram is flat, and indeed $ad - bc$ becomes $ab - ba = 0$.
- **It is** 1 **on the unit square.** With $u = e_1$ and $v = e_2$ we get 1.

The surprise of this chapter is that those three properties determine the function completely, in every dimension, and that the sign we discarded when we took the absolute value is worth keeping.



$\det[\,u \;\; v\,] = +5$ — the turn from u to v is counterclockwise

$\det[\,v \;\; u\,] = -5$ — list them the other way and the sign flips

Figure 8.1. The determinant of a 2×2 matrix is the *signed* area of the parallelogram spanned by its columns. The region is the same on both sides; what differs is the order in which the two edges are named, and the sign records that order. Discarding the sign would lose the distinction between a frame and its mirror image, which is exactly the information an orientation carries.

8.2 The three defining properties

We state them for rows rather than columns, because rows are what Chapter 3 taught us to manipulate. Proposition 8.13 will show that the choice makes no difference.

Definition 8.1 (Multilinear and alternating)**.** *Let D assign a real number to each $n \times n$ real matrix, regarded as a list of its rows $A_1, \dots, A_n$. Call D*

- multilinear *if it is linear in each row when the others are held fixed, that is,*

$$D(\dots,\ \lambda A_i + \mu A_i',\ \dots) = \lambda\, D(\dots, A_i, \dots) + \mu\, D(\dots, A_i', \dots)$$

  for every i;
- alternating *if $D(A) = 0$ whenever two rows of A are equal.*

Theorem 8.2 (Existence and uniqueness of the determinant)**.** *There is exactly one multilinear alternating function* det *on* $M_n(\mathbb{R})$ *with* $\det \mathbf{1}_n = 1$*. Moreover any multilinear alternating* D *satisfies* $D(A) = \det(A)\, D(\mathbf{1}_n)$ *for all* A*.*

We shall prove uniqueness first, in Section 8.3, because the argument is also the algorithm for computing determinants; existence, which needs a formula, comes afterwards in Section 8.4.

What the two properties already give

Proposition 8.3 (Consequences of the axioms)**.** *Let* D *be multilinear and alternating. Then, for any matrix* A*:*

(a) swapping two rows changes the sign of D*;*
(b) adding a multiple of one row to a different *row leaves* D *unchanged;*
(c) multiplying one row by λ *multiplies* D *by* λ*;*
(d) if some row is zero, or if the rows are linearly dependent, then $D(A) = 0$*.*

Proof. (a) Let u and v occupy rows i and j. Put $u + v$ in both places; the result has two equal rows, so $D = 0$. Expanding by multilinearity in row i and then in row j,

$$0 = D(\dots, u+v, \dots, u+v, \dots) = D(u,u) + D(u,v) + D(v,u) + D(v,v),$$

where we have abbreviated by showing only rows i and j. The first and last terms vanish because they have two equal rows, so $D(u,v) = -D(v,u)$.

(b) By linearity in row j,

$$D(\dots, A_i, \dots, A_j + \lambda A_i, \dots) = D(\dots, A_i, \dots, A_j, \dots) + \lambda\, D(\dots, A_i, \dots, A_i, \dots),$$

and the second term has two equal rows, hence vanishes. This is where $i \neq j$ is needed.

(c) Immediate from linearity in that row.

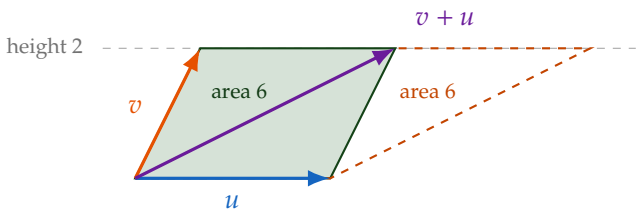
(d) A zero row gives $D = 0$ by (c) with $\lambda = 0$. If the rows are dependent then by Proposition 6.2(b) some row A_i is a combination of the others; subtracting that combination using (b) replaces A_i by the zero row without changing D. □

Key Idea

Parts (a), (b) and (c) say precisely how the three elementary row operations of Definition 3.3 act on the determinant. Since Gaussian elimination is nothing but a sequence of those operations, we can already compute any determinant, and we

have not yet had to say what one is.

Figure 8.2 shows why part (b) costs nothing: replacing v by $v + u$ slides the far edge of the parallelogram along a line parallel to u, so the base and the height, and therefore the area, are untouched.



same base, same height, same area

which is why row operations of type (R3) do not change the determinant

Figure 8.2. Proposition 8.3(b) drawn. Replacing v by $v+u$ slides the far edge of the parallelogram along the dashed line without changing the base or the height, so the area is untouched. Elimination is a sequence of such slides, which is why it can be used to compute determinants at the cost of an elimination rather than the cost of a formula.

8.3 Computing determinants, and uniqueness

Lemma 8.4 (Triangular matrices)**.** *Let D be multilinear and alternating with $D(\mathbf{1}_n) = 1$, and let U be upper triangular. Then $D(U) = u_{11}u_{22}\cdots u_{nn}$.*

Proof. Suppose first that every $u_{ii} \neq 0$. Working from the bottom row upwards, use operations of type (b) to clear every entry above each diagonal entry; this changes nothing and leaves the diagonal matrix with entries u_{ii}. Now apply (c) once per row to pull out each u_{ii}, leaving $D(\mathbf{1}_n) = 1$. Hence $D(U) = \prod_i u_{ii}$.

If instead $u_{ii} = 0$ for some i, look at rows $i, i+1, \ldots, n$. Row j of an upper triangular matrix has zeros in columns $1, \ldots, j-1$, so row j lies in span $\{e_j, \ldots, e_n\}$; for $j \geq i+1$ that is already inside span $\{e_{i+1}, \ldots, e_n\}$, and row i lies in the same span because its diagonal entry vanishes, so its first possibly nonzero entry is in column $i+1$. So all $n-i+1$ of these rows lie in a span of $n-i$ vectors and are dependent by Theorem 6.14(a). Hence $D(U) = 0$ by Proposition 8.3(d), and the product is zero too. □

Proof of uniqueness in Theorem 8.2. Let D be multilinear and alternating. Run the elimination algorithm of Chapter 3 on A, using only row swaps and operations of type

(R3). It produces an upper triangular U; suppose s swaps were used along the way. By Proposition 8.3(a) and (b),

$$D(A) = (-1)^s\, D(U) = (-1)^s\, u_{11} \cdots u_{nn}\; D(\mathbf{1}_n),$$

the last step by the argument of Lemma 8.4, which used only the axioms and therefore produces $D(\mathbf{1}_n)$ rather than 1 when D is not normalised. So D is determined by its value on the identity. Taking $D(\mathbf{1}_n) = 1$ pins it down completely, and the general statement $D(A) = \det(A) D(\mathbf{1}_n)$ falls out of the same display. □

Theorem 8.5 (Determinant by elimination)**.** *If Gaussian elimination reduces A to an upper triangular U using s row swaps and no scalings, then*

$$\det A = (-1)^s\, u_{11} u_{22} \cdots u_{nn},$$

the product of the pivots up to sign. The cost is that of one elimination, about $\frac{2}{3}n^3$ operations.

Proof. This is the display in the proof of uniqueness above, read with $D = \det$. That argument showed $D(A) = (-1)^s u_{11} \cdots u_{nn}\, D(\mathbf{1}_n)$ for any multilinear alternating D, and $\det \mathbf{1}_n = 1$ by Theorem 8.2. The cost is Proposition 3.16, since the elimination is the same one. □

Example 8.6 (A 4×4 determinant)**.** *Let*

$$A = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{pmatrix}.$$

Eliminate downwards with no swaps. The multipliers are $\frac{1}{2}$, then $\frac{2}{3}$, then $\frac{3}{4}$, and the pivots come out as

$$2, \qquad 2 - \tfrac{1}{2} = \tfrac{3}{2}, \qquad 2 - \tfrac{2}{3} = \tfrac{4}{3}, \qquad 2 - \tfrac{3}{4} = \tfrac{5}{4}.$$

Since $s = 0$,

$$\det A = 2 \cdot \tfrac{3}{2} \cdot \tfrac{4}{3} \cdot \tfrac{5}{4} = 5,$$

the intermediate factors telescoping. The same pattern in size n gives pivots $\frac{k+1}{k}$ and determinant $n + 1$, which is a useful family to remember.

Corollary 8.7 (The test for invertibility)**.** *A is invertible if and only if* $\det A \neq 0$.

Proof. By Theorem 4.16, A is invertible exactly when it has n pivots, and by Theorem 8.5 the determinant is $\pm$ the product of the pivots, which is nonzero exactly then. □

Insight

Compare the two ways to compute a determinant. Elimination costs about $\frac{2}{3}n^3$ operations. The permutation formula of the next section costs $n!$ terms. For $n = 20$ that is the difference between about 5,000 operations and about 2×10^{18}: a few microseconds against several decades. Every formula in this chapter that expresses a determinant as a sum over permutations or a recursion over minors is a tool for proving theorems, not for computing numbers, and the distinction between the two uses of a formula is one of the more valuable habits this subject teaches.

8.4 The permutation formula, and existence

To produce a function satisfying the axioms we need permutations.

Definition 8.8 (Permutations and sign)**.** *Let S_n be the set of bijections $\sigma : \{1, \dots, n\} \to \{1, \dots, n\}$. An* inversion *of σ is a pair $i < j$ with $\sigma(i) > \sigma(j)$; write* $\operatorname{inv}(\sigma)$ *for the number of them and define the* sign $\operatorname{sgn}(\sigma) = (-1)^{\operatorname{inv}(\sigma)}$*. A* transposition *is a permutation exchanging two elements and fixing the rest.*

Lemma 8.9. *Composing with a transposition reverses the sign:* $\operatorname{sgn}(\tau\sigma) = -\operatorname{sgn}(\sigma)$ *for every transposition τ. Consequently* sgn *is multiplicative and* $\operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}(\sigma)$.

Proof. Take first an *adjacent* transposition, exchanging the values in positions k and $k+1$. Every pair other than $\{k, k+1\}$ keeps its relative order, and that one pair changes from inverted to not or the other way, so $^{-1}$ changes by exactly 1 and the sign flips.

A general transposition exchanging positions i and j with $j = i + k$ can be built from adjacent ones: move the entry at position i rightwards past k neighbours, then move the displaced entry leftwards past $k-1$ neighbours. That is $2k-1$ adjacent transpositions, an odd number, so the sign flips once again.

Multiplicativity follows by writing each permutation as a product of transpositions, which is always possible, and $\operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}(\sigma)$ because $\operatorname{sgn}(\sigma)\operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}(\mathrm{id}) = 1$ and signs are ± 1. □

Theorem 8.10 (Leibniz formula)**.** *The function*

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma)\, a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)}$$

is multilinear, alternating and equal to 1 *on* $\mathbf{1}_n$*. Hence the determinant exists, completing Theorem 8.2.*

Proof. *Multilinear.* Each term contains exactly one factor from each row, so as a function of row i with the others fixed, every term is linear in the entries $a_{i1}, \ldots, a_{in}$, and a sum of linear functions is linear.

Normalised. If $A = \mathbf{1}_n$ then $a_{i\sigma(i)} = 0$ unless $\sigma(i) = i$ for every i, so only $\sigma = \mathrm{id}$ contributes, giving $\operatorname{sgn}(\mathrm{id}) \cdot 1 = 1$.

Alternating. Suppose rows p and q are equal, and let τ be the transposition exchanging p and q. Pair each σ with $\sigma\tau$; this pairs up all of S_n into disjoint couples, since $\tau^2 = \mathrm{id}$ and $\sigma\tau \neq \sigma$. The two members of a couple contribute

$$\operatorname{sgn}(\sigma) \prod_i a_{i\sigma(i)} \quad \text{and} \quad \operatorname{sgn}(\sigma\tau) \prod_i a_{i\,\sigma\tau(i)},$$

and the two products are equal, because rows p and q are identical so exchanging which of them supplies which factor changes nothing. Their signs are opposite by Lemma 8.9, so each couple contributes zero and the whole sum vanishes. □

Example 8.11 (Small cases)**.** *For $n = 2$ there are two permutations and $\det\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) = ad - bc$, which is where we came in. For $n = 3$ there are six, giving*

$$\det A = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33},$$

the rule of Sarrus: three products down the rightward diagonals minus three down the leftward ones.

Caution

Sarrus is a 3×3 accident and does not generalise. A 4×4 determinant has 24 terms, not 8, and no arrangement of diagonals produces them. Writing down the eight "diagonal" products of a 4×4 array and combining them with signs is one of the most common errors in the subject, and it gives an answer that is not merely wrong but not even multilinear. For $n \geq 4$, eliminate.

8.5 The multiplication theorem

Theorem 8.12 (Determinants multiply)**.** *For all* $A, B \in M_n(\mathbb{R})$, $\det(AB) = \det(A)\det(B)$.

Proof. Suppose first that B is not invertible. By Theorem 4.16 there is some $x \neq \mathbf{0}$ with $Bx = \mathbf{0}$, and then $(AB)x = \mathbf{0}$ too, so AB is not invertible either. Both sides are therefore zero by Corollary 8.7.

Now let B be fixed and invertible, and define $D(A) = \det(AB)$. By Proposition 4.4, row i of AB is $\sum_s a_{is} \cdot (\text{row } s \text{ of } B)$, which depends linearly on row i of A and on nothing else in A. So D is multilinear in the rows of A. It is alternating: if rows i and j of A are equal then so are rows i and j of AB, and det is alternating. By the last clause of Theorem 8.2,

$$D(A) = \det(A)\, D(\mathbf{1}_n) = \det(A)\det(B). \qquad \square$$

Proposition 8.13 (Transpose, inverse, similarity)**.** *For every* $A \in M_n(\mathbb{R})$*:*

(a) $\det(A^t) = \det(A)$*, so every statement about rows in this chapter holds for columns;*

(b) if A *is invertible then* $\det(A^{-1}) = 1/\det(A)$*;*

(c) similar matrices have equal determinants.

Proof. (a) In the Leibniz formula for A^t the term of σ is $\operatorname{sgn}(\sigma)\prod_i a_{\sigma(i)\,i}$. Re-indexing the product by $j = \sigma(i)$ turns it into $\prod_j a_{j\,\sigma^{-1}(j)}$, and $\operatorname{sgn}(\sigma) = \operatorname{sgn}(\sigma^{-1})$ by Lemma 8.9. As σ runs over S_n so does σ^{-1}, so the two sums have exactly the same terms.

(b) Apply Theorem 8.12 to $AA^{-1} = \mathbf{1}_n$ and use $\det \mathbf{1}_n = 1$.

(c) $\det(P^{-1}AP) = \det(P^{-1})\det(A)\det(P) = \det(A)$ by (b) and commutativity of real multiplication. $\square$

Key Idea

Part (c) discharges the promise made in Chapter 7. Since all matrices representing one operator T in different bases are similar, they share a determinant, and we may write $\det T$ without naming a basis. Together with the rank and the trace, this gives three genuine invariants of an operator; Chapter 9 will add the eigenvalues, of which the trace and determinant turn out to be the sum and the product.

Example 8.14 (The reflection again). *The reflection of Example 7.25 had matrices*

$$[T]_E = \tfrac{1}{5}\begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix} \qquad \text{and} \qquad [T]_B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Their determinants are $\frac{1}{25}\big(3(-3) - 4\cdot 4\big) = \frac{-25}{25} = -1$ *and* $1\cdot(-1) - 0 = -1$. *They agree, as Proposition 8.13(c) requires, and the common value* -1 *says something the arrays did not: this map preserves area and reverses orientation, which is what a mirror does. A rotation, by contrast, has determinant* $+1$.

8.6 Cofactors, the adjugate and Cramer's rule

Definition 8.15 (Minors and cofactors). *For* $A \in M_n(\mathbb{R})$ *let* $A(i \mid j)$ *denote the* $(n-1)\times(n-1)$ *matrix obtained by deleting row* i *and column* j. *The* (i,j) minor *is* $\det A(i \mid j)$ *and the* cofactor *is* $C_{ij} = (-1)^{i+j} \det A(i \mid j)$.

Theorem 8.16 (Laplace expansion). *For any fixed row* i, *and likewise for any fixed column* j,

$$\det A = \sum_{j=1}^{n} a_{ij}\, C_{ij} \qquad \text{and} \qquad \det A = \sum_{i=1}^{n} a_{ij}\, C_{ij}.$$

Sketch. Group the terms of the Leibniz formula according to the value $j = \sigma(i)$. The terms with $\sigma(i) = j$ contain the common factor a_{ij}, and what remains is a sum over the bijections from $\{1,\dots,n\}\setminus\{i\}$ to $\{1,\dots,n\}\setminus\{j\}$, which is the Leibniz formula for $A(i \mid j)$ once the indices are relabelled. Tracking what the relabelling does to inversions produces the factor $(-1)^{i+j}$. The column statement then follows from Proposition 8.13(a). □

Proposition 8.17 (Adjugate and inverse). *Let* $\operatorname{adj} A$ *be the transpose of the matrix of cofactors, so* $(\operatorname{adj} A)_{ij} = C_{ji}$. *Then*

$$A\,\operatorname{adj} A = \operatorname{adj} A \cdot A = (\det A)\,\mathbf{1}_n,$$

and if $\det A \neq 0$ *then* $A^{-1} = \dfrac{1}{\det A}\operatorname{adj} A$.

Proof. The (i,i) entry of $A \operatorname{adj} A$ is $\sum_j a_{ij}C_{ij} = \det A$ by Theorem 8.16. For $i \neq k$, the (i,k) entry is $\sum_j a_{ij}C_{kj}$, which is the Laplace expansion along row k of the matrix obtained

from A by replacing row k with a copy of row i. That matrix has two equal rows, so its determinant is zero. Hence $A \operatorname{adj} A = (\det A)\mathbf{1}_n$.

For the other order, apply what has just been proved to A^t. Deleting row j and column i from A^t gives $A(i \mid j)^t$, so the cofactors of A^t are the cofactors of A transposed, and $\operatorname{adj}(A^t) = (\operatorname{adj} A)^t$ by Proposition 8.13(a). Transposing $A^t \operatorname{adj}(A^t) = (\det A^t)\mathbf{1}_n$ therefore gives $\operatorname{adj} A \cdot A = (\det A)\mathbf{1}_n$. □

Corollary 8.18 (Cramer's rule)**.** *If* $\det A \neq 0$*, the unique solution of* $Ax = b$ *has*

$$x_j = \frac{\det A_j}{\det A},$$

where A_j *is* A *with its* j*th column replaced by* b*.*

Proof. From $x = A^{-1}b = (\det A)^{-1} \operatorname{adj}(A)\, b$, the jth entry is $(\det A)^{-1} \sum_i C_{ij} b_i$, and that sum is the Laplace expansion of $\det A_j$ along its jth column. □

Caution

Cramer's rule is a beautiful formula and a terrible algorithm. It needs $n + 1$ determinants; even computing each by elimination that is $O(n^4)$ work against the $O(n^3)$ of simply solving the system, and evaluating them by cofactors instead is astronomically worse. It is also numerically unstable, since the two determinants can both be tiny while their ratio is not. The same applies to $A^{-1} = \operatorname{adj}(A)/\det A$: use it to prove that the inverse depends smoothly on the entries, which it does and which matters, and never to compute one.

8.7 Volume

The determinant has been built from three axioms and computed by elimination, and it is time to say what it measures. Figure 8.3 is the statement in the plane: the unit square has area 1 and its image has area $|\det A|$, so every region is scaled by that one number.

Signed volume in $\mathbb{R}^n$, however one chooses to define it rigorously, has the three properties of Section 8.2: it is linear in each edge of the parallelepiped when the others are held fixed, it vanishes when two edges coincide because the figure is then flat, and it is 1 on the unit cube. By Theorem 8.2 there is only one such function. Hence:

Theorem 8.19 (Determinant as a volume factor)**.** *The signed volume of the parallelepiped spanned by* $v_1, \dots, v_n$ *is* $\det[\, v_1 \ \cdots \ v_n \,]$*. Consequently, for any region* $S \subseteq \mathbb{R}^n$

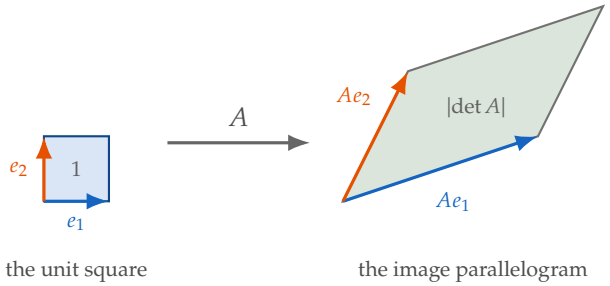


Figure 8.3. The determinant as a volume factor. The unit square has area 1 and its image has area $|\det A|$, so every region has its area multiplied by that number. Because scaling factors compose by multiplication, this is also the geometric content of Theorem 8.12: doing B and then A multiplies volume by $|\det B|$ and then by $|\det A|$.

with a volume,

$$\mathrm{Vol}\big(A(S)\big) = |\det A| \cdot \mathrm{Vol}(S),$$

and A preserves orientation exactly when $\det A > 0$.

With the determinant read as an area, Proposition 8.3(b) needs no algebra at all, and Figure 8.2 can now be read in the other direction: it is because the slide preserves area that elimination may be used to evaluate a determinant. The proposition adds a row where the picture slides a column, which is the same statement because $\det A = \det A^t$.

Insight

This is the reason the determinant appears in the change-of-variables formula of multivariable calculus. A smooth change of coordinates is not linear, but it is approximately linear near each point, and the local linear approximation is the Jacobian matrix; the factor $|\det J|$ in the integral is exactly Theorem 8.19 applied to that approximation, one small region at a time. The same quantity reappears in probability, where the density of a transformed random variable carries a factor $|\det J|^{-1}$, and in machine learning, where normalising flows are built precisely from transformations whose Jacobian determinant is cheap to evaluate. That design constraint, which looks arbitrary when first met, is this theorem.

Example 8.20 (Orientation and degeneracy). *The matrix* $\left(\begin{smallmatrix} 3 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ *has determinant* 5*: it multiplies area by* 5 *and preserves orientation, which is Figure 8.1 on the left. Swapping its columns gives determinant* -5*: the same area factor, orientation reversed. The matrix*

$\left(\begin{smallmatrix}1&2\\2&4\end{smallmatrix}\right)$ *has determinant* 0*, and indeed it collapses the plane onto the line spanned by* $(1,2)$*, which has area zero; this is Corollary 8.7 and Example 2.7 seen from a third side.*

Open Problem 8.21 (The Hadamard matrix conjecture)**.** *Among all* $n \times n$ *matrices with entries in* $\{-1,+1\}$*, how large can the determinant be? Hadamard's inequality gives* $|\det A| \le n^{n/2}$*, since each row has length* $\sqrt{n}$ *and the determinant is at most the product of the row lengths. Equality requires the rows to be orthogonal, and such a matrix is called a* Hadamard matrix*. Orthogonality of* ± 1 *rows forces* $n = 1$*,* $n = 2$*, or* n *divisible by* 4*.*
Does a Hadamard matrix exist for every n divisible by 4?
What is known. *Constructions cover a great many orders: Sylvester's doubling gives every power of two, Paley's construction covers* $q+1$ *and* $2(q+1)$ *for prime powers* q*, and combinations of these plus sporadic computer searches have settled every multiple of four below* 668*. The smallest open order was* 428 *until a matrix was found in 2004; as of this writing* 668 *is the smallest order for which no Hadamard matrix is known [333]. The conjecture has been open since 1893.*
Where the difficulty sits. *The condition is* $HH^t = n\mathbf{1}$*, which is* $\binom{n}{2}$ *quadratic equations in* n^2 *sign variables. Counting says solutions should be plentiful, and they are, for the orders we can reach; but the constructions are all algebraic, built from finite fields or groups, and the multiples of four are not organised by any algebraic structure.* There is no reason to expect a single construction to cover them all, and no reason to expect a counterexample either.
Where a reader might start. *Build the* 4×4 *and* 8×8 *Hadamard matrices by Sylvester's doubling* $H \mapsto \left(\begin{smallmatrix}H & H\\ H & -H\end{smallmatrix}\right)$ *and check* $|\det| = n^{n/2}$ *directly. Then try* $n = 12$*, where doubling does not reach, and discover why a construction rather than a search is needed. These matrices reappear in Chapter 29 as optimal codes and in Chapter 24 as cheap approximations to random orthogonal matrices.*

Exercises

Exercise 8.22. *Compute each determinant, by elimination where that is quicker.*

(a) $\left(\begin{smallmatrix}4&7\\2&6\end{smallmatrix}\right)$

(b) $\left(\begin{smallmatrix}1&2&3\\0&4&5\\0&0&6\end{smallmatrix}\right)$

(c) $\left(\begin{smallmatrix}2&0&1\\1&3&2\\1&1&1\end{smallmatrix}\right)$

(d) $\left(\begin{smallmatrix}0&1&2\\1&0&3\\2&3&0\end{smallmatrix}\right)$

Exercise 8.23. *Let $A \in M_n(\mathbb{R})$ with $\det A = 3$. Find $\det(2A)$, $\det(A^2)$, $\det(A^t A)$ and $\det(A^{-1})$, each in terms of n where necessary. Explain why $\det(2A) \neq 2\det A$ in general, given that the determinant is multilinear.*

Exercise 8.24. *Show that*

$$\det\begin{pmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{pmatrix} = (b-a)(c-a)(c-b).$$

Hint: subtract row 1 from the others, then factor. *Deduce that the matrix is invertible exactly when a, b, c are distinct, and say what that means about fitting a quadratic through three points.*

Exercise 8.25. *Let A and D be square. Prove that* $\det\left(\begin{smallmatrix} A & B \\ 0 & D \end{smallmatrix}\right) = \det A \cdot \det D$. Hint: eliminate within the blocks, or fix D and check the axioms in the rows of A. *Then give a 4×4 example with 2×2 blocks showing that* $\det\left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right) = \det A \det D - \det B \det C$ *is* false *in general.* Hint: arrange for two rows of the big matrix to be equal while $\det B = \det C = 0$.

Exercise 8.26. *Suppose $Q^t Q = \mathbf{1}_n$. Show that $\det Q = \pm 1$, and interpret both cases geometrically using Theorem 8.19. Which case does the reflection of Example 8.14 fall into?*

Exercise 8.27. *Solve $2x + y = 5$, $x + 3y = 10$ twice: once by Cramer's rule and once by elimination. Then count the arithmetic operations each method used, and extrapolate to $n = 10$.*

Exercise 8.28. *Prove from the axioms alone, without the Leibniz formula, that if D is multilinear and alternating then $D(A) = 0$ whenever A has two* proportional *rows. Then show that the multilinear alternating functions on $M_n(\mathbb{R})$ form a vector space of dimension 1, and name a basis for it.* Hint: Theorem 8.2 does all the work.

Exercise 8.29. *Let T_n be the $n \times n$ matrix with 2 on the diagonal, 1 immediately above and below, and 0 elsewhere, as in Example 8.6. Prove that $\det T_n = 2\det T_{n-1} - \det T_{n-2}$ by Laplace expansion along the first row, and deduce $\det T_n = n + 1$.*

Exercise 8.30. *Compute $\operatorname{sgn}(\sigma)$ for the permutation of $\{1,2,3,4,5\}$ sending $(1,2,3,4,5)$ to $(3,5,1,4,2)$, by counting inversions and again by writing it as a product of transpositions. Then write down the corresponding term of the Leibniz formula.*

References and further reading

The history is well documented and the order of events is genuinely surprising, so it is worth reading about: determinants were studied for a century and a half before anyone thought the array itself deserved a name. Our axiomatic treatment, defining the determinant by multilinearity, alternation and normalisation and only then producing a formula, is the modern standard and goes back to Weierstrass and Kronecker. Axler [37] famously postpones determinants to the last chapter and develops the spectral theory without them, which is worth seeing precisely because it shows how much of the spectral theory does not depend on this chapter. Hefferon [307] and Austin [35] both take the geometric route, building from signed area, and Austin is particularly good on orientation. Banica [51], our baseline, treats determinants briskly and puts them to work immediately on the characteristic polynomial, which is where Chapter 9 will need them. For the numerical side, and in particular for why no serious code computes a determinant when it can avoid one, Trefethen and Bau [607] is again the reference; their observation that $\det A$ is a poor measure of how close A is to singular, the condition number being the right one, is worth internalising early.